

A SCIENCE & ENGINEERING TALK

# The Engineering of Play

How games grew up — and where they are going — told through the engineering that made each leap possible.

**Soutam Rajbhar**

Department of Chemical Engineering  
IIT Kharagpur



WHY THIS MATTERS

# Gaming is the **biggest** entertainment business on Earth.

Bigger than movies and music put together. And it isn't really an art industry. It's an engineering one.

~\$200B

MONEY MADE EACH YEAR

3.3B+

PEOPLE WHO PLAY

1 in 3

PEOPLE ON THE PLANET PLAY

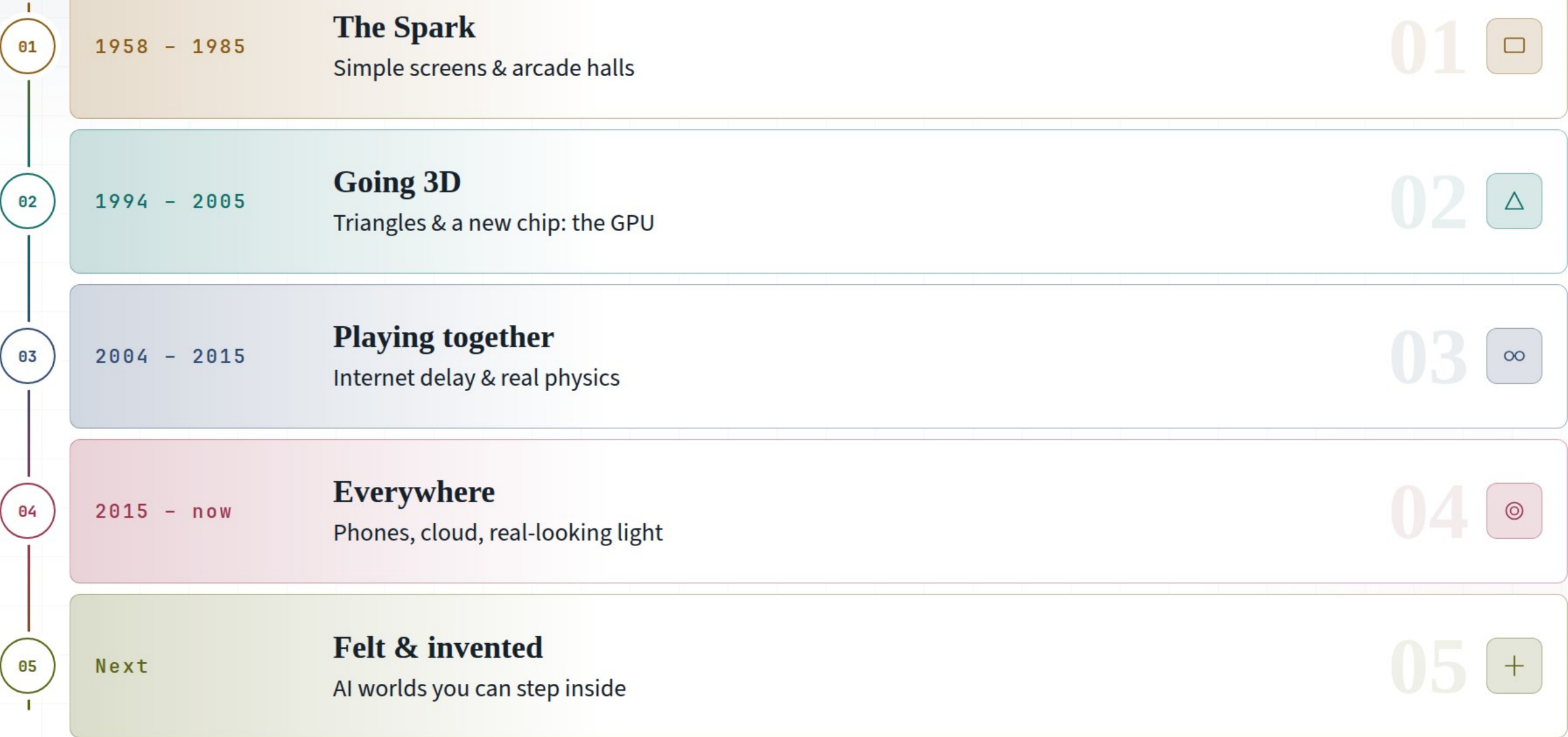
60–240

NEW PICTURES DRAWN EVERY  
SECOND

Every second, your computer does maths for millions of points — shape, bounce, light — and paints a new picture before you can blink. That's the story we're following today.

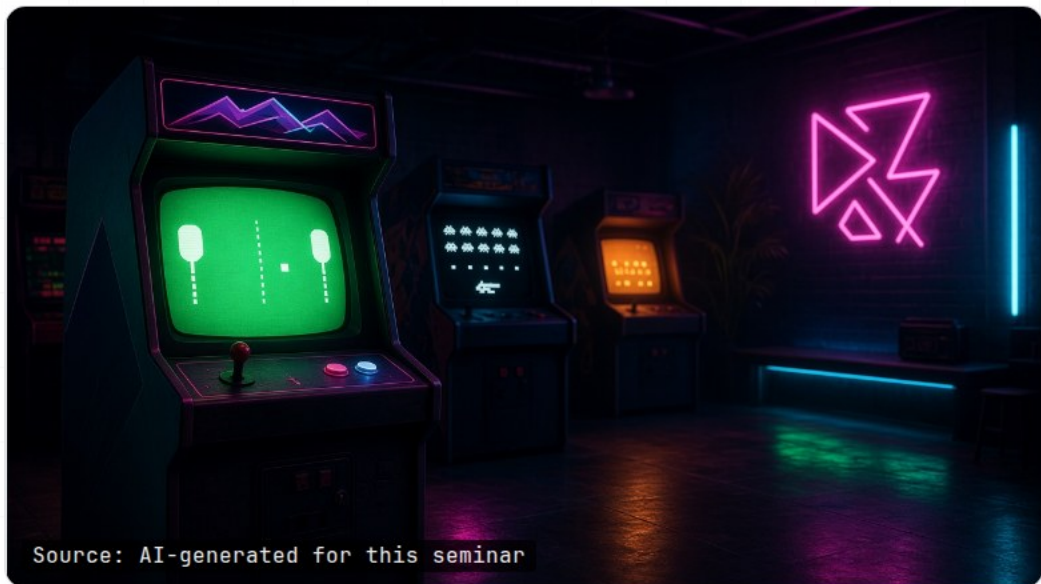
# Five big leaps. One story.

Each leap happened because engineers solved a hard problem. Here's the map for today.



# The Spark: making a screen talk back

Before fancy pictures, the hard question was simple: can a machine react to a person right now?



## The games

*Tennis for Two* (1958) on a lab screen, *Pong* (1972), then *Space Invaders* and packed arcade halls.

## How it was built

Early machines were wired by hand. There was often no “brain chip” yet — the wiring *itself* was the game.

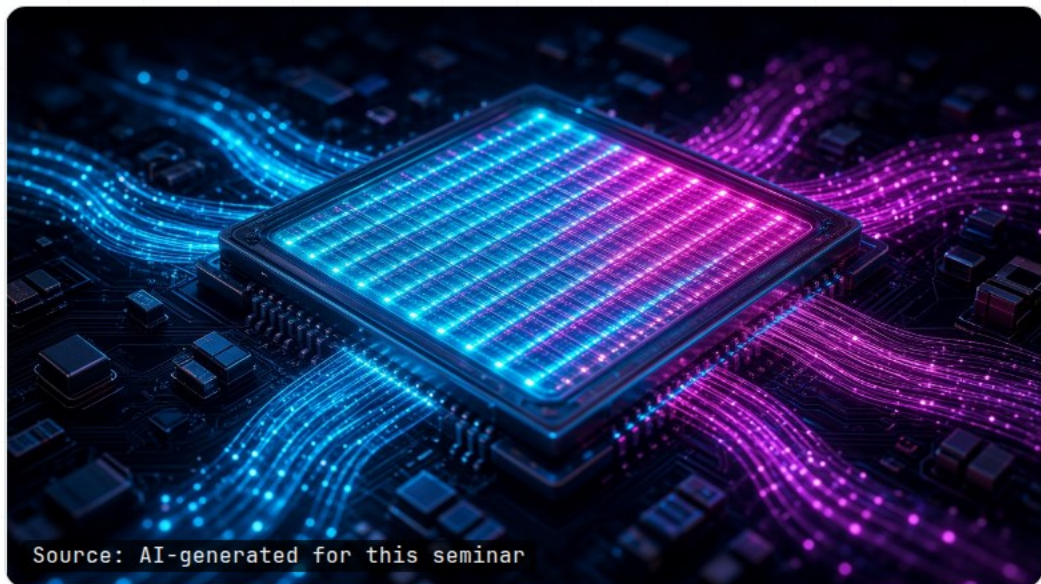
## The big change

Cheap computer chips arrived. A game became *software* — the same machine could play many different games.

**Takeaway:** once games were software, progress stopped being about new wires and started being about faster computers.

# Games go 3D — and a new chip is born: the GPU

To turn flat pictures into worlds, a computer had to do the same maths millions of times — very fast.



Source: AI-generated for this seminar

## The problem

A 3D world is made of millions of tiny triangles. Every picture, the computer must move them, light them, and colour every dot. A normal chip (the CPU) does this one step at a time — too slow for smooth play.

## The answer

A new chip built just for pictures: the **Graphics Processing Unit**. It has many small workers that colour lots of dots *at the same time*. NVIDIA's 1999 GeForce 256 was sold as the first of these.

PlayStation

Nintendo 64

Quake

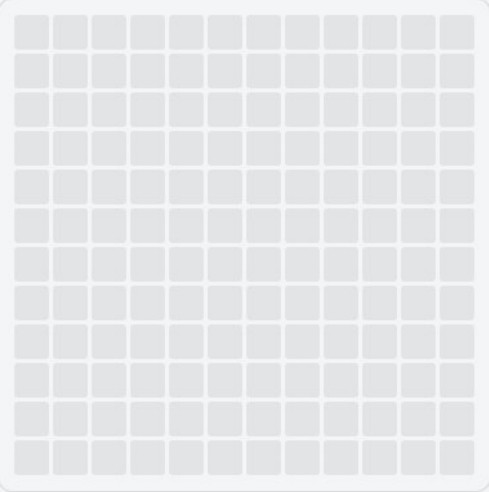
GeForce 256

One idea matters more than anything else today: do the same small job on thousands of things at once. Let's watch it happen. →

TRY IT YOURSELF • TAP THE BUTTONS

# Why a GPU? Many hands at once.

Both chips colour the same 144-dot picture. The CPU has 4 strong workers. The GPU has 48 smaller ones. Watch the clock.



shaded

0/144

time

0.00s

CPU • 4 cores

GPU • 48 cores

## CPU • 4 cores

Few workers. Each one is smart and can do many kinds of jobs — like 4 master chefs in a kitchen.

## GPU • 48 cores

Many workers. Each one is simple and repeats one job — like 48 line cooks all chopping vegetables at once.

If we gave the CPU 48 cores, would it become a GPU? No. More chefs still aren't a factory line. A real GPU has thousands of these simple cores. That same trick now trains AI.

# Playing together: **delay**, physics & huge worlds

Home internet turned games into places we share. That created new headaches for engineers.

01



## Fighting delay

A signal can take about 40 milliseconds to cross the world. Games guess your next move, then correct it — so play still feels instant.

01

02



## Real physics

Games solve bounce, crash, and falling bodies 60 times a second — the same kind of maths used in engineering simulations.

02

03



## Huge worlds

Games like *World of Warcraft* keep tens of thousands of players in sync across many computers at once.

03

The shift: a game stopped being “a program on your computer” and became a live system serving millions of people at once.

# Games go everywhere

From a phone in every pocket to light that looks almost real.

01



## In every pocket

Most gaming now happens on phones. A modern phone already has a strong picture chip — a console in your hand.

01

02



## Games in the cloud

The game can run on a faraway computer and send you video — like Netflix, except your button press has to travel there and back in a blink.

02

03



## Tracing light

New chips follow rays of light through a scene — real reflections and shadows. Movie computers used to spend hours on one frame. Games now do it live.

03

04



## Esports

Competitive games fill stadiums. That needs rock-solid servers, fair play, and timing accurate to a fraction of a picture.

04

# Gaming tech quietly **rebuilt the world**

Tools made so we could have fun ended up powering some of today's most serious work.

01



## GPUs → the AI boom

The same “many hands” maths that colours pixels is what trains AI. Chatbots and image models run on chips that started life in games.

01

02



## Game tools → film & design

The same software that builds games now draws movie sets, car showrooms, and buildings — live, not overnight.

02

03



## Game physics → factories

Robots and self-driving cars practise in game-like worlds. Factories test a virtual copy of a plant before they build the real one.

03

**The point:** chasing a fun, hard problem — drawing a world 60 times a second — gave every other field a new set of tools.

# Worlds that are **invented**, not just drawn

The next leap: the computer doesn't only paint the world. It starts to imagine it.

## 01 AI fills in the picture

Games already draw a rough picture, then let AI add the extra detail — more sharpness for less work. NVIDIA calls one version of this DLSS.

01

## 02 Worlds made on the spot

Research models can now invent playable game pictures in real time. Imagine levels, characters, and quests made just for you.

02

## 03 Characters that talk back

Chat-style AI can give game characters real conversations — they reply to what you actually say, not a fixed script.

03

## 04 Endless maps + AI

Older tricks already built whole galaxies from one seed number. Pair that with AI and you get huge worlds that still feel surprising.

04

# From watching a screen to a world you **step inside**

The next race isn't only prettier pictures. It's making the world feel close enough to touch.



## Headsets & mixed reality

A headset must draw two sharp pictures, many times a second, with almost no delay after you move your head. Miss that, and people feel sick.

## Feeling the world

Motors in controllers and suits let you feel weight, texture, and kickback — not just see it.

## Play with almost no hardware

As networks get faster, the heavy work can live in the cloud — a rich world streamed to thin, cheap glasses.

## Playing by thought

Early brain-computer links already let some people move a cursor by thinking. That's a far-off idea of how we might one day play.

# The same story, over and over

01

## A wild wish

Someone wants a rich world that reacts right now.

01

02

## A clever leap

Engineers answer with a new chip or a smart trick — the GPU, delay-hiding, tracing light, AI drawing.

02

03

## It leaves the game

That trick becomes a tool for everyone — AI, film, robots, and science.

03

Games are where we practise building **worlds that react in real time** — and that practise keeps handing the rest of engineering its next tools.

# Key takeaways

1

## **Gaming is an engineering story**

Bigger than film and music put together — and every leap was won by engineers, not only artists.

2

## **The GPU is the hero chip**

Built to colour dots side by side, it now trains almost every big AI model. Fun hardware became world-changing hardware.

3

## **Game tricks keep leaving the game**

Picture chips, physics, online play, and AI drawing all leaked out to power AI, film, robots, and science.

4

## **The future is invented and felt**

AI will imagine worlds as you play them, and headsets, touch, and networks will make those worlds feel close.

WHERE THE FACTS COME FROM

# References

A starting list if you want to read more after today.

01 Newzoo. Global Games Market Report — how much money and how many players · 2024

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03 NVIDIA. DLSS — AI that adds extra detail to a game picture · 2020–

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05 Valve Developer Community. Source Multiplayer Networking — hiding internet delay

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07 Google Research. GameNGen — a neural net that can run a playable game · 2024

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02 NVIDIA. GeForce 256 — sold as “the world’s first GPU”, plus later chip history · 1999–

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04 Akenine-Möller, Haines & Hoffman. Real-Time Rendering (4th ed.) — how live pictures are drawn · 2018

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06 Google DeepMind. AlphaGo & AlphaStar — AI that learned by playing games · 2016–19

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08 Cooper et al., Nature. Foldit — a game that helped fold a real protein · 2010

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Numbers are rounded for scale. Every picture carries its own source line. Pictures in this talk are original AI-generated illustrations made for this seminar — not photos of real products.

ONE LAST THOUGHT

# The next leap in technology won't be announced in a lab — it'll ship in a game.

Dots → triangles → many hands at once → AI → worlds you can step inside.  
From *Pong* on a lab screen to places we may one day walk into — every leap  
was an engineering leap. Watch what gamers play next. It's a preview of what  
everyone else will build with.

Thank you — questions welcome

Soutam Rajbhar · Chemical Engineering, IIT Kharagpur